

1 THE ODD SYMMETRIC THEOREM

1.1 Statement

Work over a field of characteristic $\neq 2$ (the campaign uses $\mathbb{F}$; the theorem is characteristic-free away from 2). Let $S = \mathbb{F}[s_1, \dots, s_6]$. For each of the fifteen perfect matchings J of the six indices, let

$$I_J := (s_a + s_b : \{a, b\} \in J) \subseteq S,$$

the ideal of the 3-plane V_J on which the variables pair off with opposite signs. Let $e_k = e_k(s_1, \dots, s_6)$ denote the elementary symmetric polynomials.

Theorem (Odd Symmetric). $\bigcap_{J \in \mathcal{J}} I_J = (e_1, e_3, e_5)$. Moreover (e_1, e_3, e_5) is a regular sequence; the quotient $S/(e_1, e_3, e_5)$ is a reduced complete intersection of dimension 3 and degree 15, whose irreducible components are exactly the fifteen planes V_J .

No even elementary symmetric polynomial lies in any I_J (e_2 restricts to $-(x^2+y^2+z^2) \neq 0$ on every V_J), so the odd ideal is exact — neither generator can be dropped, none can be added.

1.2 Proof

Step 1 (: the odd symmetrics vanish on every pairing plane). Fix J and parametrize V_J by $(x, -x, y, -y, z, -z)$. For odd k , split the k -subsets of the six slots by how they meet the three pairs. A subset containing a full pair contributes the factor $x \cdot (-x)$ times a term whose complementary part is summed symmetrically over the remaining slots; grouping these by the complement, each group's sum is the elementary symmetric of $\{y, -y, z, -z\}$ -type sets of odd residual size, and pairs of terms differing by one sign-flip cancel. A subset meeting each chosen pair in exactly one slot carries an odd number of sign choices in total, and the involution flipping the sign of one fixed selected pair matches terms in cancelling couples. Concretely: $e_1 = x - x + y - y + z - z = 0$; $e_3 = 0$ and $e_5 = 0$ by the same cancellation (for e_5 , the leave-one-out products come in $\pm$ pairs). Hence $e_1, e_3, e_5 \in I_J$ for all J , i.e. $(e_1, e_3, e_5) \subseteq \bigcap_{J \in \mathcal{J}} I_J$.

Step 2 (the varieties agree). Let $s = (s_1, \dots, s_6) \in \mathbb{F}^6$ satisfy $e_1(s) = e_3(s) = e_5(s) = 0$. Then

$$(t - s_1)(t - s_2) = t^2 + e_1 t + e_2 t^2 + e_3$$

is a polynomial in t^2 . Since $\text{char} \neq 2$, its root multiset is invariant under $t \mapsto -t$, and the multiplicity of the root 0 is even. Therefore the multiset $\{s_1, \dots, s_6\}$ has the form $\{s, -s, s, -s, s, -s\}$, and one may choose a perfect matching J of the indices pairing each value with an opposite one: s_i lies on V_J . Hence $V(e_1, e_3, e_5) = \bigcup_{J \in \mathcal{J}} V_J$ set-theoretically.

Step 3 (regular sequence, degree, reducedness). By Step 2 the variety of the 3-generated ideal (e_1, e_3, e_5) has dimension 3, so the ideal has height 3 and e_1, e_3, e_5 is a regular sequence. The quotient is a complete intersection, hence Cohen–Macaulay and unmixed, of degree $1 \cdot 3 \cdot 5 = 15$ by Bézout. Its support consists of the fifteen pairwise-distinct linear 3-planes V_J , each of degree 1; unmixedness forces every component to appear, and the degree count $15 = \sum (\text{multiplicities})$ over fifteen components forces every multiplicity to equal 1. A generically reduced Cohen–Macaulay scheme (Serre's $R_1 + S_1$) is reduced, so (e_1, e_3, e_5) is a radical ideal. Consequently

$$(e, e, e) = I(_J V_J) = _J I_J.$$

1.3 Remarks

1. **Numerical confirmation.** Over F the Hilbert functions of both sides agree in degrees 1–7: 1, 6, 22, 61, 142, 292, 547 (probe log `PROBE_V57_ODD_SYMMETRIC_v1.log`; per-leg memberships of e, e verified for all fifteen matchings). The identity itself is proven for all degrees by the argument above and carries no parameters.
2. **Symmetry.** Both sides are S -stable; the fifteen matchings form a single S -orbit with stabilizer of order 48, and the theorem identifies the ideal of that orbit of 3-planes with the span of the odd half of the fundamental invariants — a complete intersection presentation of a subspace arrangement, with Bézout number equal to the orbit size.
3. **Frobenius compatibility (char 3).** $e_k(s)^3 = e_k(s^3)$: in characteristic 3 the elementary symmetric polynomials commute with the Frobenius endomorphism, since cubing is additive. This is the property that makes the theorem the correct closed form of the campaign’s bone: the object is stable under the very map that generates the tower.
4. **Role in the campaign.** With $m_v = (s^{\{3_v\}}, \dots, s^{\{3_v\}})$, the Degtyarev–Shimada saturation statement for the Fermat tower is equivalent to $_J (I_J + m_v) = (e, e, e) + m_v$ for all v . The theorem above is the m -free core of that statement; the tower inclusion is treated in the campaign documents (V58).

Part of the Sofa Theorem repository: github.com/tretoef-estrella · tretoef@gmail.com